

# Jiwoo Kim

ROBOTICS RESEARCHER · 3D GEOMETRIC PERCEPTION

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Sensor-agnostic geometric perception for safe and scalable robotics

## Summary

I study 3D geometric perception for safe and scalable robotic systems. My research focuses on sensor-agnostic methods that reason about geometry across heterogeneous sensing modalities, with an emphasis on robustness and uncertainty quantification.

## Highlights

- Open-source research: Doppler Correspondence [★ 46] and Learning Point Cloud Geometry [★ 120].
- Best Student Paper Finalist at Robotics: Science and Systems (RSS) 2025; Outstanding Paper Award at the Institute of Control, Robotics and Systems (ICROS) 2024.
- Research interests: geometric perception, deep learning, and uncertainty quantification.

## Education

### Korea Advanced Institute of Science and Technology (KAIST)

PH.D. STUDIES IN MECHANICAL ENGINEERING

Mar. 2026 - Present

### Ulsan National Institute of Science and Technology (UNIST)

M.S. IN MECHANICAL ENGINEERING

Mar. 2024 - Feb. 2026

- GPA: 3.72 / 4.5

### Sejong University

B.S. IN AEROSPACE ENGINEERING

Mar. 2016 - Feb. 2020

- GPA: 3.95 / 4.5

## Research Project Experience

### Lunar Landing Project

WITH KOREA AEROSPACE RESEARCH INSTITUTE (KARI)

2026

- LiDAR-inertial SLAM for lunar environments.
- Hazard detection and avoidance (HDA).

### AI Model Compression

WITH AGENCY FOR DEFENSE DEVELOPMENT (ADD)

## Honors & Awards

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|------|---------------------------------------------------------------------------------------------------------------------------|---------------|
| 2027 | <b>Workshop Organizer</b> , Workshop on Spatial Memory, IEEE International Conference on Robotics and Automation (ICRA)   | International |
| 2026 | <b>1st Place, Preliminary Round</b> , AI Grand Prix, Anduril                                                              | International |
| 2026 | <b>2nd Place, Main Competition Round 1</b> , AI Grand Prix, Anduril; advanced to the next round                           | International |
| 2025 | <b>Best Student Paper Finalist</b> , Robotics: Science and Systems (RSS); one of 6 finalists among 163 accepted papers    | International |
| 2024 | <b>Outstanding Paper Award</b> , Institute of Control, Robotics and Systems (ICROS); ranked 1st among 518 accepted papers | Domestic      |

# Publications

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## INTERNATIONAL JOURNAL

1. Park HyoungHo, Hongro Jang, Seunghwan Kim, Junhee Lee, **Jiwoo Kim**, and Hyondong Oh, “Semantic-GSL: Semantically Guided Gas Source Localization With Vision-Language Models,” *IEEE Robotics and Automation Letters (RA-L)*, 2026.

## INTERNATIONAL CONFERENCE

1. **Jiwoo Kim**, Jinwoo Lee, Woojae Shin, Giseop Kim, and Hyondong Oh, “Learning the Underlying Geometry as a Statistical Manifold: Theory and Practice,” *Robotics: Science and Systems (RSS)*, 2026. [📄 ★ 120]
2. **Jiwoo Kim**, Geunsik Bae, Changseung Kim, Jinwoo Lee, Woojae Shin, and Hyondong Oh, “Doppler Correspondence: Non-Iterative Scan Matching With Doppler Velocity-Based Correspondence,” *Robotics: Science and Systems (RSS)*, 2025. [📄 ★ 46]

## INTERNATIONAL WORKSHOP

1. **Jiwoo Kim**, Jinwoo Lee, Woojae Shin, Giseop Kim, and Hyondong Oh, “Self-supervised Geometry Reasoning for LiDAR Simultaneous Localization and Mapping,” *Robotics: Science and Systems (RSS) Workshop*, 2026.
2. Park HyoungHo, Hongro Jang, Seunghwan Kim, Junhee Lee, **Jiwoo Kim**, and Hyondong Oh, “Semantic-GSL: Semantically Guided Gas Source Localization With Vision-Language Models,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop*, 2025.

## DOMESTIC CONFERENCE

1. Changseung Kim, **Jiwoo Kim**, and Hyondong Oh, “EKF-Based Radar-Inertial Odometry With Online Temporal Calibration,” *Institute of Control, Robotics and Systems (ICROS)*, 2024.

## MANUSCRIPTS IN PREPARATION

1. **Uncertainty Quantification for Neural Networks**. First author. In preparation; target venue: *International Conference on Learning Representations (ICLR)*, 2027.
2. **Learning-Based Multi-View Point Cloud Registration**. First author. In preparation; target venue: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2027.
3. **Latent Analysis of Geometric Foundation Models**. First author. In preparation; target venue: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2027.
4. **Learning-Based Trajectory Planning**. Second author. In preparation; target venue: *IEEE Robotics and Automation Letters (RA-L)*, 2027.

# Academic Service

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## IEEE Robotics and Automation Letters (RA-L)

PEER REVIEWER

2025 - 2026

- 2026: Koopman Kalman filtering; LiDAR odometry.
- 2025: Legged-robot odometry; monocular SLAM in dynamic scenes.

## Robotics: Science and Systems (RSS) Workshop

PAPER REVIEWER

2026